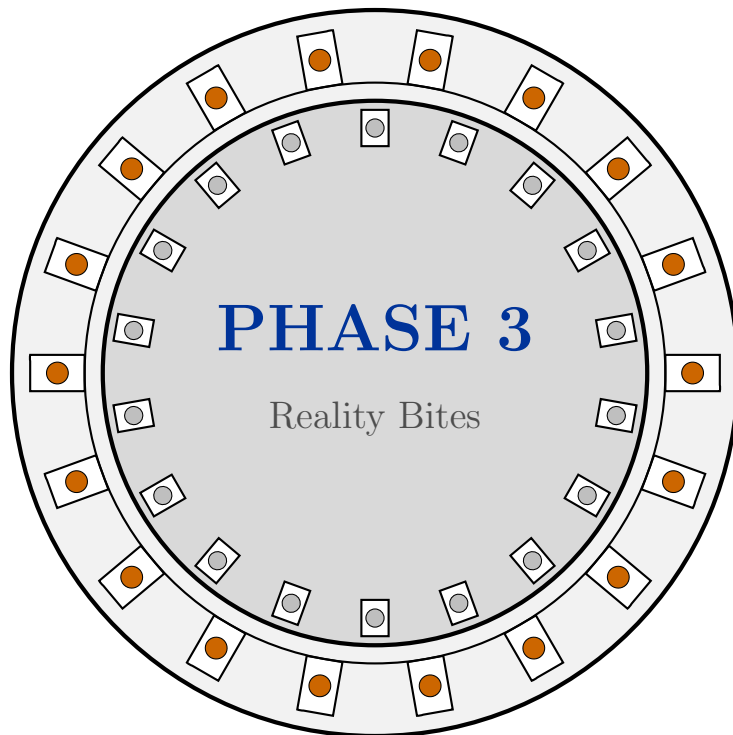


PHASE 3: REAL-WORLD CONSTRUCTION & FLAWS

Descending into Physical Reality

An Absolute Zero-to-Mastery Roadmap

Conquering Slots, Winding Factors, Space Harmonics, Cogging, and Crawling.



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Introduction: The Death of the Ideal Machine

If we wanted to build the mathematical ideal motor from Phase 1 and 2, we would take a perfectly smooth steel cylinder (the stator), and we would literally glue our copper wires directly onto the smooth inner surface.

Why don't we do this? 1. **Mechanical Failure:** The magnetic forces inside a 50 kW motor are incredibly violent. If wires were just glued to the surface, the spinning magnetic tornado would rip them right off the steel and shred them to pieces. 2. **Air Gap Size:** The magnetic flux has to jump from the stator to the rotor. Air is a terrible conductor of magnetism (it has incredibly high *reluctance*). If we put thick copper wires on the surface, we would have to leave a huge air gap between the stator and the rotor to fit the wires. A huge air gap requires a massive, inefficient magnetizing current (I_m) just to push the flux across.

The Mental Sandbox (Visualize This)

The Engineering Solution: Instead of putting wires on the surface, we cut deep trenches (**Slots**) into the steel stator. We bury the copper wires safely inside these trenches.

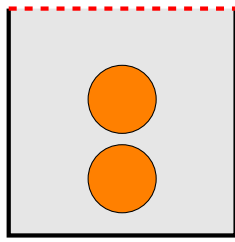
Now, the steel "teeth" between the slots can be brought extremely close to the rotor. The air gap becomes microscopic (just 1 or 2 millimeters). The wires are protected from mechanical ripping, and the magnetic flux easily jumps across the tiny air gap.

But placing wires into discrete, separate slots destroys our perfectly smooth, continuous mathematical sine wave. It turns our smooth magnetic field into a staircase. This single physical decision is the root cause of every problem we will study in Phase 3.

Block 1: Stator Slot Geometry (The 2023 Q1b Question)

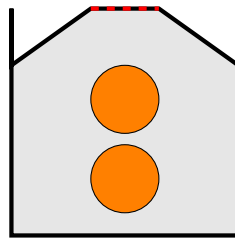
When we cut slots into the stator, we must choose a shape. This is a classic exam question: "*Why stators in a 3-phase induction motor are usually provided with semi-closed type slots?*"

Let's visually examine the three types of slots.



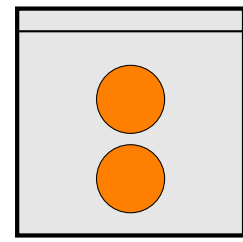
Open Slot

Wide opening.
Easy to insert coils.
High reluctance.



Semi-Closed Slot

Narrow opening.
Harder to insert.
Excellent compromise.



Closed Slot

No opening.
Wire must be threaded.
High leakage flux.

Figure 1: Cross-sectional view of the three primary stator slot designs.

The Verdict on Semi-Closed Slots

For small and medium rating induction motors, we almost exclusively use **Semi-Closed Slots**. Here is the exact reasoning you must write in the exam:

1. **Reduced Air-Gap Reluctance:** The opening (the red dashed line) is the "air" part of the magnetic circuit. Air has high magnetic resistance (reluctance). By making the opening narrow, we maximize the amount of steel facing the rotor. More steel = lower reluctance.
2. **Improved Power Factor:** Because the reluctance is lower, the motor requires significantly less Magnetizing Current (I_m) to push the flux across the air gap. Since I_m is highly inductive, reducing it drastically improves the overall Power Factor ($\cos \phi$) of the motor.
3. **Reduced Tooth Pulsation Loss:** A wide open slot creates sudden, violent changes in magnetic flux density as the rotor spins past it. A semi-closed slot smooths out the inner stator surface, reducing high-frequency magnetic noise and surface iron losses.

The Trade-off: You cannot drop a pre-formed, rigid coil into a semi-closed slot. Factory workers (or machines) must feed the individual copper wires through the narrow slit one by one.

Block 2: Winding Architecture - The Pitch Factor (K_p)

Now we have our semi-closed slots. We need to wind our coils into them.

A single "coil" consists of two "coil sides" connected by an end-turn. One coil side goes down Slot A, and the return side comes back through Slot B. The distance between Slot A and

Slot B is called the **Coil Span** or **Coil Pitch**.

The Mental Sandbox (Visualize This)

The Ideal Scenario (Full Pitch): A magnetic pole spans 180° electrical. The absolute maximum voltage is induced if we place Coil Side 1 exactly under the dead-center of the North pole, and Coil Side 2 exactly under the dead-center of the South pole.

This means the Coil Span is exactly 180° . The voltage induced in side 1 (E_1) and side 2 (E_2) will be perfectly in phase with each other. They add together arithmetically: $E_{total} = E_1 + E_2$.

The Reality (Short Pitch / Chording): In reality, we deliberately make the coil span *shorter* than 180° by an angle α . This is called a **Short-Pitched** or **Chorded** winding.

Why on earth would we intentionally reduce our voltage? 1. **Copper Savings:** A shorter coil means shorter end-turns. This saves massive amounts of expensive copper and reduces the physical bulk of the machine. 2. **Reduced Resistance:** Less copper means lower winding resistance (R_1), which reduces I^2R heat losses and increases efficiency. 3. **Harmonic Elimination:** This is the killer feature. By carefully choosing the angle α , we can mathematically cancel out specific, destructive space harmonics (like the 5th and 7th).

Semester Exam Trap

2024 Q1d: "Find an expression for pitch factor for a short pitched winding." (Note: The paper had a typo, spelling it 'peach factor'). You must be able to derive $K_p = \cos(\alpha/2)$ using a clear phasor diagram.

Derivation of Pitch Factor (K_p)

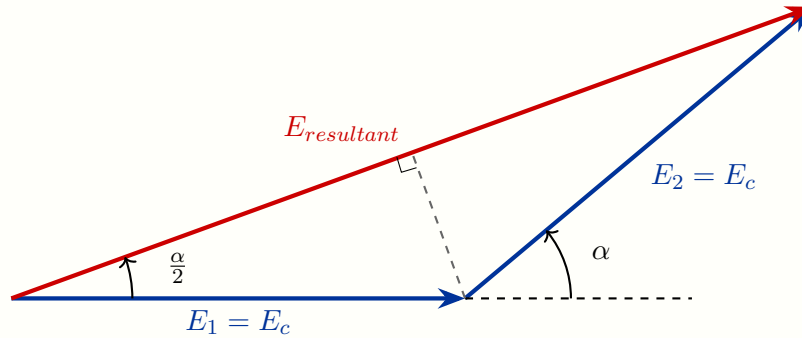
Definition: Pitch factor is the ratio of the actual phasor sum of the coil EMFs to the arithmetic sum (if it had been full-pitched).

$$K_p = \frac{\text{Phasor Sum of EMFs}}{\text{Arithmetic Sum of EMFs}} \quad (1)$$

Let E_c be the EMF induced in one single coil side. If the coil was full-pitched, the two sides are perfectly in phase.

$$E_{arithmetic} = E_c + E_c = 2E_c \quad (2)$$

Because the coil is short-pitched by an angle α , the EMF in the second side is phase-shifted by α with respect to the first side. We must add them using vector (phasor) addition.



Look at the right-angled triangle formed by the origin, the midpoint of the resultant (M), and the end of E_1 (A). The hypotenuse is E_c . The adjacent side is exactly half of the resultant EMF ($E_{res}/2$). The angle is $\alpha/2$.

Using basic trigonometry ($\cos \theta = \text{adjacent/hypotenuse}$):

$$\cos\left(\frac{\alpha}{2}\right) = \frac{E_{res}/2}{E_c} \quad (3)$$

Rearranging to solve for the actual resultant EMF:

$$E_{res} = 2E_c \cos\left(\frac{\alpha}{2}\right) \quad (4)$$

Now, plug this into the definition of Pitch Factor:

$$K_p = \frac{E_{res}}{E_{arithmetic}} = \frac{2E_c \cos\left(\frac{\alpha}{2}\right)}{2E_c} \quad (5)$$

$$\boxed{K_p = \cos\left(\frac{\alpha}{2}\right)} \quad (6)$$

Block 3: Winding Architecture - The

Distribution Factor (K_d)

We have our shortened coil. Now, what do we do with all the coils that belong to Phase R?

The Mental Sandbox (Visualize This)

The Concentrated Nightmare: If we have 10 coils belonging to Phase R for a single magnetic pole, the simplest thing to do is stuff all 10 coils into ONE giant slot. This is a *Concentrated Winding*.

Why is this terrible? 1. A single slot deep enough to hold 10 coils would severely weaken the mechanical integrity of the steel core. 2. 10 coils packed together generate massive, concentrated heat. The inner wires would melt because they can't dissipate heat. 3. It produces a harsh, rectangular block of magnetic flux, totally ruining our smooth sine wave ideal.

The Reality (Distributed Winding): Instead of one slot, we use multiple adjacent slots (e.g., 3 slots) per pole per phase. We place a few coils in slot 1, a few in slot 2, and a few in slot 3.

This spreads the heat beautifully. It strengthens the core. Most importantly, it smooths out the stepped magnetic wave, making it look much closer to a pure sine wave.

However, because slot 1, slot 2, and slot 3 are at different physical angles, the voltages induced in them will peak at slightly different times. When we wire them in series, we cannot add their voltages arithmetically ($10V + 10V + 10V \neq 30V$). We must add them vectorially, which results in a slightly lower total voltage.

Semester Exam Trap

2022 Q1d & 2025 Q1d: "What do you mean by distributed winding? Find an expression for distribution factor." You must draw the phasor arc and prove the geometric chord formula.

Derivation of Distribution Factor (K_d)

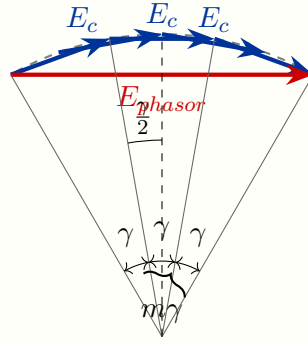
Definition: The Distribution Factor (or Breadth Factor) is the ratio of the phasor sum of the coil EMFs to their arithmetic sum.

$$K_d = \frac{\text{Phasor Sum of distributed coils}}{\text{Arithmetic Sum}} \quad (7)$$

Let:

- m = number of slots per pole per phase.
- γ = electrical angle between adjacent slots (slot pitch).
- E_c = EMF induced in one single slot.

If the winding was concentrated, all m coils would be in phase: $E_{arithmetic} = m \times E_c$. Because they are distributed, they form a polygon of phasors, each shifted by angle γ . In geometry, m consecutive vectors of equal length shifted by a constant angle form the chords of a circumscribing circle.



Look at the geometry. Let the radius of the circle be R . Take one single coil voltage (E_c). It is a chord subtending angle γ at the center. Dropping a perpendicular bisects it into a right triangle with angle $\gamma/2$ and opposite side $E_c/2$.

$$\sin\left(\frac{\gamma}{2}\right) = \frac{E_c/2}{R} \implies E_c = 2R \sin\left(\frac{\gamma}{2}\right) \quad (8)$$

The arithmetic sum of m coils is:

$$E_{arithmetic} = m \times 2R \sin\left(\frac{\gamma}{2}\right) \quad (9)$$

Now look at the red Resultant vector (E_{phasor}). It is a single massive chord subtending the total angle $m\gamma$ at the center. Dropping a perpendicular bisects it into a right triangle with angle $m\gamma/2$ and opposite side $E_{phasor}/2$.

$$\sin\left(\frac{m\gamma}{2}\right) = \frac{E_{phasor}/2}{R} \implies E_{phasor} = 2R \sin\left(\frac{m\gamma}{2}\right) \quad (10)$$

Divide the two to find K_d :

$$K_d = \frac{2R \sin\left(\frac{m\gamma}{2}\right)}{m \times 2R \sin\left(\frac{\gamma}{2}\right)} \quad (11)$$

$$K_d = \frac{\sin\left(\frac{m\gamma}{2}\right)}{m \sin\left(\frac{\gamma}{2}\right)} \quad (12)$$

The Fundamental Winding Factor (K_w)

Combining both realities (short-pitching and distributing), the actual voltage induced in the machine is lower than the theoretical mathematical ideal. We define the total Winding Factor K_w as:

$$K_w = K_p \times K_d \quad (13)$$

Actual induced voltage per phase: $E_{actual} = 4.44f\Phi N_{ph} \times K_w$.

Block 4: The Birth of Space Harmonics (The $6k \pm 1$ Rule)

We have distributed our windings, but they are still sitting in discrete slots. Because the copper is grouped in slots rather than spread uniformly like butter, the Magnetomotive Force (MMF) produced by the stator is NOT a smooth sine wave. It is a staircase.

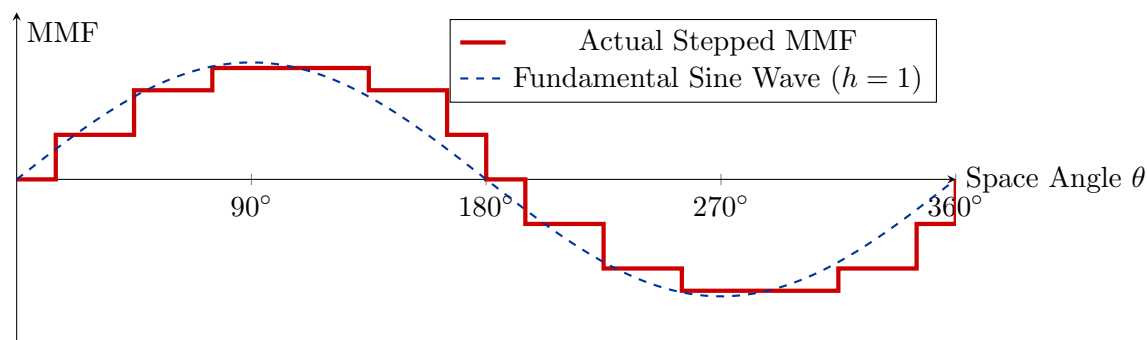


Figure 2: The stepped MMF wave created by placing coils in discrete slots.

According to the mathematical wizardry of Jean-Baptiste Joseph Fourier, any repeating, non-sinusoidal waveform can be broken down into an infinite sum of perfect, pure sine waves of increasing frequencies. These are called **Harmonics**.

So, our single "staircase" magnetic field is actually a ghost army of hundreds of perfectly smooth sine waves, all spinning in the air gap at different speeds and in different directions!

Semester Exam Trap

2023 Q1e: "Show that a 3-phase winding carrying sinusoidal currents produces space harmonics of the order of $h = 6k \pm 1$." This requires a logical, two-step elimination proof.

Proof of the $6k \pm 1$ Harmonic Rule

Let h represent the harmonic order ($h = 1$ is the fundamental, $h = 2$ is the 2nd harmonic, etc.).

Step 1: Elimination of Even Harmonics (Half-Wave Symmetry) Look at the staircase wave. The North pole (positive half, 0° to 180°) is physically identical to the South pole (negative half, 180° to 360°), just flipped upside down. Mathematically, this is called **Half-Wave Symmetry**: $F(\theta) = -F(\theta + 180^\circ)$. According to Fourier series rules, a waveform with half-wave symmetry contains **absolutely no even harmonics**. Therefore, $h = 2, 4, 6, 8, 10 \dots$ are completely eliminated. The field only contains odd harmonics: $h = 1, 3, 5, 7, 9, 11, 13, 15 \dots$

Step 2: Elimination of Triplen Harmonics in 3-Phase Systems The remaining odd harmonics include the "triplens" (multiples of 3): $h = 3, 9, 15 \dots$. Let us analyze how the 3^{rd} harmonic behaves across the three phases (R, Y, B). In the fundamental wave ($h = 1$), the phases are displaced by 120° . For the h -th harmonic, this physical displacement angle is multiplied by h .

For the 3^{rd} harmonic ($h = 3$), the phase displacement becomes:

$$3 \times 120^\circ = 360^\circ \equiv 0^\circ \text{ (electrical)} \quad (14)$$

This is a massive revelation. It means that the 3rd harmonic magnetic field produced by Phase R, Phase Y, and Phase B are all perfectly synchronized (0° phase difference). They are **co-phasal**.

Because the motor is fed by a 3-wire, balanced 3-phase AC supply without a neutral return, Kirchhoff's Current Law demands that the sum of the instantaneous currents is zero ($i_R + i_Y + i_B = 0$). Since the 3rd harmonic fields are perfectly in phase with their currents, their instantaneous sum must also be zero! Therefore, **all triplen harmonics** ($3, 9, 15 \dots$) **perfectly cancel each other out to zero** in the air gap.

Step 3: The Survivors Take the set of odd numbers, and remove the multiples of 3. What survives?

$$h = 1, 5, 7, 11, 13, 17, 19 \dots$$

This infinite mathematical sequence can be elegantly expressed by the single formula:

$$\boxed{h = 6k \pm 1} \quad (15)$$

where k is a positive integer ($1, 2, 3 \dots$).

Block 5: The Monsters Awaken (5th and 7th Harmonics)

We have proven that our stator is generating a fundamental field ($h = 1$), a 5th harmonic field, a 7th harmonic field, an 11th, and so on. The magnitude of these harmonics drops off rapidly ($F_h = F_1/h$), so we only really care about the strongest survivors: the 5th and the 7th.

What do these ghost fields do to our rotor?

The Mental Sandbox (Visualize This)

Speed of Harmonics: A harmonic field of order h behaves exactly like a separate motor with $h \times P$ poles. Since synchronous speed is $N_s = 120f/P$, increasing the poles by h means the speed drops by h .

$$N_{s,h} = \frac{N_s}{h}$$

The 5th harmonic spins at $\frac{1}{5}N_s$. The 7th harmonic spins at $\frac{1}{7}N_s$.

Semester Exam Trap

2023 Q1f: "Show that the 5th harmonic produces backward rotating torque whereas 7th produces forward rotating torque." This requires phase sequence analysis.

Direction of Rotation (Phase Sequence Analysis): The direction an RMF spins depends entirely on its phase sequence (A-B-C spins forward, A-C-B spins backward). The fundamental sequence is A-B-C (displaced by 120° and 240°).

1. The 5th Harmonic ($h = 5$): Multiply the fundamental phase displacement by 5:

- Phase B displacement: $5 \times 120^\circ = 600^\circ$. Subtracting 360° gives a principal angle of $600 - 360 = \mathbf{240^\circ}$.
- Phase C displacement: $5 \times 240^\circ = 1200^\circ$. Subtracting multiples of 360 ($3 \times 360 = 1080$): $1200 - 1080 = \mathbf{120^\circ}$.

Look at the results. Phase B is now at 240° and Phase C is at 120° . The original sequence was A(0)-B(120)-C(240). The new sequence is **A(0)-C(120)-B(240)**. The sequence has reversed! Therefore, the 5th harmonic magnetic field spins **BACKWARD** at a speed of $-N_s/5$. It creates a parasitic braking torque that tries to slow the motor down.

2. The 7th Harmonic ($h = 7$): Multiply the fundamental phase displacement by 7:

- Phase B displacement: $7 \times 120^\circ = 840^\circ$. Subtracting 720° gives a principal angle of $840 - 720 = \mathbf{120^\circ}$.
- Phase C displacement: $7 \times 240^\circ = 1680^\circ$. Subtracting 1440° gives a principal angle of $1680 - 1440 = \mathbf{240^\circ}$.

The angles are identically 120° and 240° . The phase sequence remains **A(0)-B(120)-C(240)**. Therefore, the 7th harmonic magnetic field spins **FORWARD** at a speed of $+N_s/7$. It creates a parasitic motoring torque pushing in the same direction, but at a very slow speed.

Block 6: Crawling (The 7th Harmonic Trap)

Now we take the beautiful, smooth torque-speed curve of the fundamental wave (from Phase 2) and superimpose the tiny parasitic torque curves of the 5th and 7th harmonics.

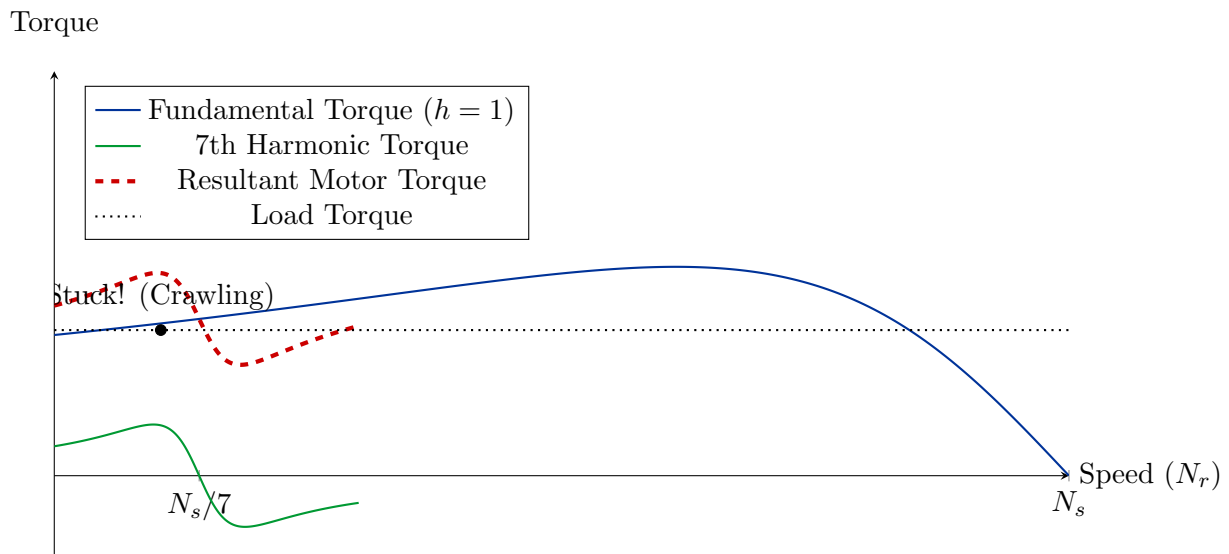


Figure 3: The 7th harmonic creates a localized "dip" or "saddle" in the total torque curve.

The Physics of Crawling

Crawling is a phenomenon where a squirrel cage induction motor starts, but refuses to accelerate to its rated speed. Instead, it gets "stuck" running stably at a very low speed (approximately one-seventh of synchronous speed, $N_s/7$), making a terrible humming noise.

Why does it happen? Look at the red dashed curve in the graph above. The 7th harmonic creates its own forward-driving torque curve that hits synchronous speed at $N_s/7$. Just past $N_s/7$, the 7th harmonic slip becomes negative, turning it into a generator, which produces a sharp **braking torque**.

This sharp drop creates a "dip" or "saddle" in the total resultant torque of the motor. If you connect a heavy mechanical load (the dotted black line) to the motor, the motor must generate more torque than the load to accelerate. At the saddle point, the motor's total torque drops *below* the load torque. The acceleration drops to zero. The motor is trapped. It will spin endlessly at $N_s/7$ and quickly burn up due to high current.

Block 7: Cogging (Magnetic Locking)

While crawling is getting stuck at low speed, **Cogging** is worse: the motor refuses to start at all. It sits at 0 RPM, locked in place like it's superglued, humming violently.

The Mental Sandbox (Visualize This)

The Magnetic Circuit Path of Least Resistance: Magnetic flux behaves like a lazy person. It will always take the easiest possible path (the path of minimum reluctance). Air is hard to cross. Iron is easy to cross.

Imagine a stator with 24 slots (and 24 iron teeth). Imagine a rotor that also happens to have exactly 24 slots (and 24 iron teeth).

When the motor is assembled, there will be positions where the 24 stator teeth align perfectly, face-to-face, with the 24 rotor teeth. In this exact position, the air gap distance is absolutely minimized. The reluctance of the magnetic circuit is at its absolute lowest point.

The Physics of Cogging

If the number of stator slots (S_1) is equal to the number of rotor slots (S_2), or an integral multiple ($S_1 = kS_2$), the rotor will strongly align its teeth with the stator teeth to maintain the state of minimum reluctance.

The magnetic flux acts like superglue between the aligned teeth. This creates a massive **Reluctance Torque** fighting against any rotational movement. If this locking reluctance torque is stronger than the starting electromagnetic torque developed by the windings, the rotor will not budge. This is called **Cogging** or Magnetic Locking.

The Engineering Fix: 1. **Never make $S_1 = S_2$.** Rotor slots are carefully chosen to be prime numbers relative to the stator slots to ensure alignment never happens across the whole rotor simultaneously. 2. **Skewing:** The rotor slots are not cut perfectly straight across the cylinder. They are twisted (skewed) slightly at an angle. Because the slot is angled, a single rotor tooth can never perfectly align with a straight stator tooth across its entire length. This completely destroys the locking force.

Phase 3 Completion Checklist

You have descended from the mathematical heavens into the gritty reality of machine construction. You have learned how engineers compromise to build real motors, and how to mathematically analyze the flaws those compromises create.

Verify your mastery of these concepts before moving to Phase 4:

- [] Do I understand why semi-closed slots are used to reduce reluctance and improve power factor?
- [] Can I sketch the phasor diagram to prove $K_p = \cos(\alpha/2)$ for a short-pitched coil?
- [] Can I sketch the phasor polygon to prove K_d for a distributed winding?
- [] Do I know the logical steps to prove that a 3-phase winding only produces $6k \pm 1$ harmonics? (Hint: Half-wave symmetry kills evens, 3-phase connection kills triplens).
- [] Can I explain to an examiner exactly why the 5th harmonic spins backward and the 7th spins forward?

- [] Do I understand the physical difference between Cogging ($S_1 = S_2$ lock) and Crawling (7th harmonic saddle)?

The Mental Sandbox (Visualize This)

The Bridge to Phase 4: We now fully understand the physical motor, its rotating fields, its speeds, and its flaws. But to an electrical engineer, a physical machine is useless unless we can calculate exactly how many Amps it will draw and how many Watts of heat it will produce.

We need to turn this heavy iron machine into a simple drawing on a piece of paper consisting only of Resistors and Inductors. In Phase 4, we will construct the **Exact Equivalent Circuit** and learn how to extract its hidden parameters using laboratory tests.